



NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

Third Semester B. Tech (CBCS) Degree Examinations

December 2025

VT2001-1 – ELECTRONIC DEVICES AND TECHNOLOGY

(For VLSI)

Duration: 3 Hours

Max. Marks: 100

Note:

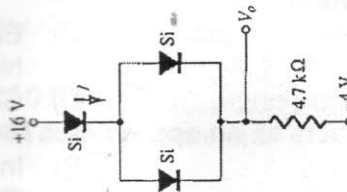
Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

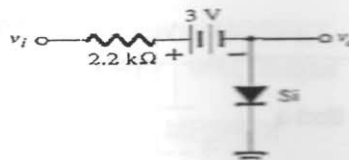
PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- Which fabrication process is used to grow a thin oxide layer on silicon?
A) Ion implantation
B) Thermal oxidation
C) Metallization
D) Photolithography
- The built-in voltage of a PN junction depends on
A) Applied bias
B) Only Temperature
C) Doping concentrations
D) Junction area
- For the Circuit shown assuming Silicon Diode with a $V_f(\text{Si}) = 0.7\text{V}$, the value of Voltage V_o is.



- A) 16V
C) -4V
- B) 14.6V
D) 15.3V
4. Einstein's relation connects which two parameters?
A) Diffusion coefficient and mobility
C) Drift current and diffusion current
- B) Mobility and conductivity
D) Voltage and current
5. Which of the following is responsible for reverse breakdown in a heavily doped diode?
A) Avalanche breakdown
C) Thermal breakdown
- B) Zener breakdown
D) Punch-through
6. Diffusion capacitance in a PN junction is dominant when the diode is:
A) Reverse biased
C) Forward biased
- B) Unbiased
D) At breakdown
7. Which optoelectronic device emits light when forward biased?
A) Photodiode
C) Solar cell
- B) LDR
D) LED
8. For the Circuit shown assuming Silicon Diode with a $V_T = 0.7$, what is the maximum Output voltage that can be got if the input is a 8V sine wave.



- A) 5V
C) 3V

- B) $0.7V$
D) $-0.7V$

9. The gate voltage required for the conduction of an n-channel enhanced mode MOSFET having a threshold voltage of 2V is
 A) 0 V B) 1 V
 C) 2 V D) More than 2 V
10. Transit frequency in MOSFET is directly proportional to
 A) Gate capacitance B) Source resistance
 C) Transconductance D) Channel length
11. The enhancement mode MOSFET is used for _____ in digital CMOS circuits.
 A) Amplification B) Switching
 C) Rectification D) Clipping
12. A MOSFET carries a drain current of 1 mA with $V_{DS} = 0.5$ V in saturation. If V_{DS} rises to 1 V and $\lambda = 0.1 \text{ V}^{-1}$. What is the device output impedance?
 A) 9.6k Ω B) 96k Ω
 C) 10k Ω D) 10.42k Ω
13. FinFETs are preferred over planar MOSFETs due to:
 A) Better scalability B) Higher leakage current
 C) Lower mobility D) Simpler fabrication
14. The high electron mobility in HEMTs is due to:
 A) Heavy doping of the channel B) Use of a wide bandgap material
 C) Separation of carriers from dopants D) Low threshold voltage
15. The transconductance of FET depends upon
 A) Drain supply B) The type of FET
 C) Gate to source voltage D) Gate current
16. Velocity saturation occurs when:
 A) Gate voltage exceeds threshold Voltage B) Electric field in the channel becomes very high
 C) Drain current becomes zero D) Channel length increases
17. In which region a transistor acts as an open switch?
 A) Cut off region B) Inversion region
 C) Active region D) Saturation region
18. In the Eber's-Moll model, the emitter and collector currents are expressed as functions of
 A) Base current only B) Junction voltages
 C) External resistances D) Load impedance
19. For a BJT, $I_C = 100$ mA and $I_E = 100.2$ mA. The value of β is
 A) 500 B) 200
 C) 1 D) 50
20. Transit frequency f_T of a BJT is defined as the frequency at which
 A) The base current becomes zero B) The current gain drops to unity
 C) The collector current reaches maximum D) The transistor enters saturation

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit - I

1. a) With a neat diagram, explain the process of growing Single crystal Silicon.
 b) Using the Diodes 1N4148 having a $V_f = 0.5$ V Implement the Circuit blocks given in Fig. 1(b) if the input signal has a frequency of 10kHz. Choose Capacitor of 1 μ F.

Marks	BT*	CO*	PO*
8	L*2	1	1
8	L3	2	1

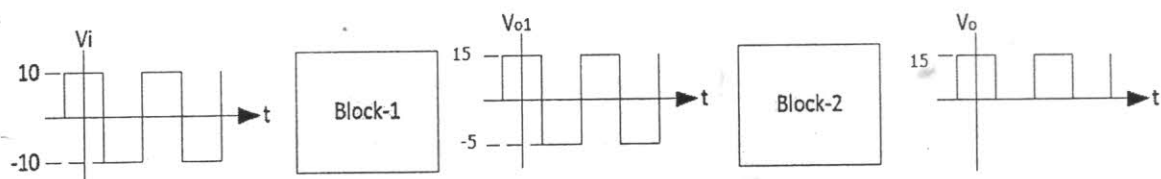


Fig. 1(b)

2. a) With a neat diagram, briefly explain the process of vapor phase epitaxy. 6 L2 1 1
- b) A silicon pn junction employs $N_A = 2 \times 10^{16} \text{ cm}^{-3}$ and $N_D = 4 \times 10^{16} \text{ cm}^{-3}$. Determine the built-in potential at room temperature. Given $n_i = 1.5 \times 10^{10} / \text{cm}^3$. 4 L3 2 1
- c) Find the resistivity of i) Intrinsic silicon and ii) p-type silicon with, $N_A = 10^{16} / \text{cm}^3$. Use $n_i = 1.5 \times 10^{10} / \text{cm}^3$ and assume that for intrinsic silicon, $\mu_n = 1350 \text{ cm}^2 / \text{V.s}$, $\mu_p = 480 \text{ cm}^2 / \text{V.s}$ and for the doped silicon, $\mu_n = 1110 \text{ cm}^2 / \text{V.s}$, $\mu_p = 400 \text{ cm}^2 / \text{V.s}$ 6 L3 1 1
3. a) With a neat diagram explain the process of thermal oxidation. 8 L2 1 1
- b) Design a circuit that limits a signal defined by the equation $8\sin(1000\pi t)$ to $\pm 4\text{V}$. 8 L3 2 1

Unit – II

4. a) With respect to MOS transistors explain
(i) Transconductance
(ii) Channel Length Modulation 6 L2 3 1
- b) An NMOS device operating in saturation must provide a transconductance of $1/(50\Omega)$. Assuming $\mu_n C_{ox} = 200 \mu\text{A/V}^2$, $\lambda = 0$, and $V_{TH} = 0.4\text{V}$,
i. Determine W/L if $I_D = 0.5 \text{ mA}$.
ii. Determine W/L if $V_{GS} - V_{TH} = 0.5\text{V}$
iii. Determine I_D if $V_{GS} - V_{TH} = 0.5\text{V}$ 6 L3 4 1
- c) A MOSFET is biased at a drain current of 0.5 mA . If $\mu_n C_{ox} = 200 \mu\text{A/V}^2$, $W/L = 10$, and $\lambda = 0.1\text{V}^{-1}$, calculate its small-signal parameters. 4 L3 4 1
5. a) With neat diagrams, explain the formation of depletion region and channel in MOS transistor. 6 L2 3 1
- b) With a neat model and device symbol with capacitances, explain the high-frequency model of n-channel MOSFET. 6 L2 4 1
- c) In the circuit of Fig. 5(c), $\mu_n C_{ox} = 200 \mu\text{A/V}^2$, $V_{TH} = 0.4\text{V}$, $W/L = 20/0.18$ and $\lambda = 0$. What value of V_B places the transistor at the edge of saturation. Also determine the small signal parameters.

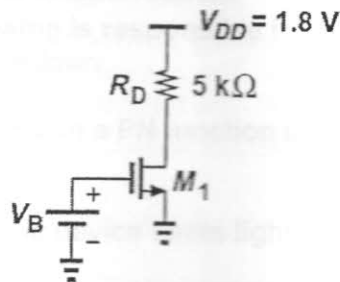


Fig. 5(c)

4 L3 4 1

6. a) Draw a neat diagram detailing the structure of an n-channel MOSFET. Explain its operation giving the plots of transfer and output characteristics. 6 L2 3 1
- b) With relevant sketches explain the fabrication of GaAs MESFET. 10 L2 4 1

Unit – III

7. a) Draw a neat sketch of high frequency model of a BJT and derive the expression for transit frequency 6 L2 5 1

- b) Draw the small-signal model for the circuit shown in Fig. 7(b).

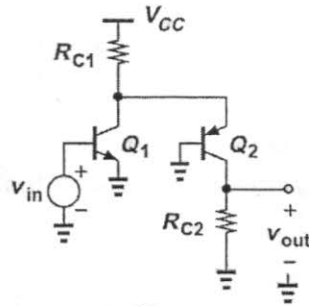


Fig. 7(b)

- | | | | | | |
|-------|--|---|----|---|---|
| c) | With relevant sketches explain the working of PNP Transistor in Active Mode. | 4 | L3 | 5 | 1 |
| | | 6 | L2 | 5 | 1 |
| 8. a) | With relevant sketches explain the working of NPN Transistor in Active Mode. | 8 | L2 | 5 | 1 |
| b) | Draw the small signal model for BJT assuming $V_A \neq \infty$. Derive the expressions for small signal parameters. | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
